

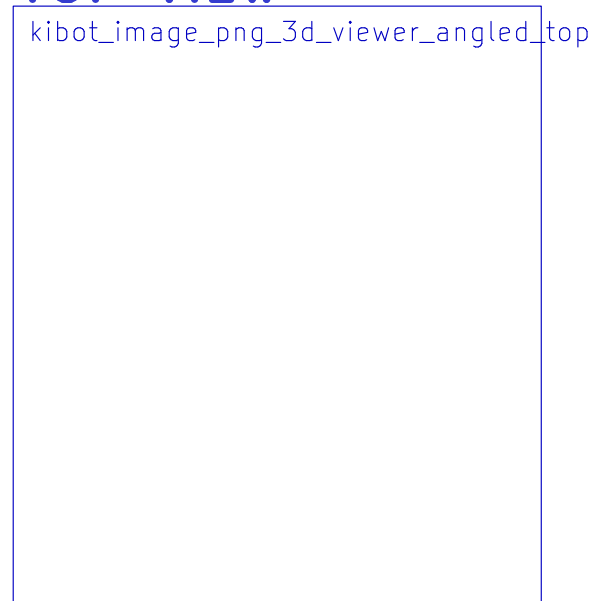
INPUT PROTECTION

VARIANT: DRAFT

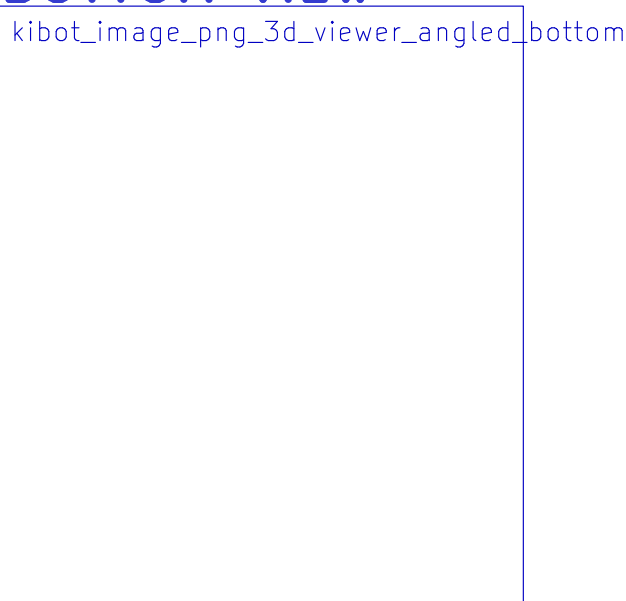
REV

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TOP VIEW



BOTTOM VIEW



NOTES

COMMENT

NOT FITTED COMPONENTS MARKED AS **X**

DRAFT – EARLY STAGE SCHEMATIC CAPTURE
PRELIMINARY – LATE STAGE SCHEMATIC CAPTURE
CHECKED – PCB LAYOUT CHECKED, CONTACT ENGINEER IF MISTAKE FOUND
RELEASED – READY FOR PRODUCTION

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

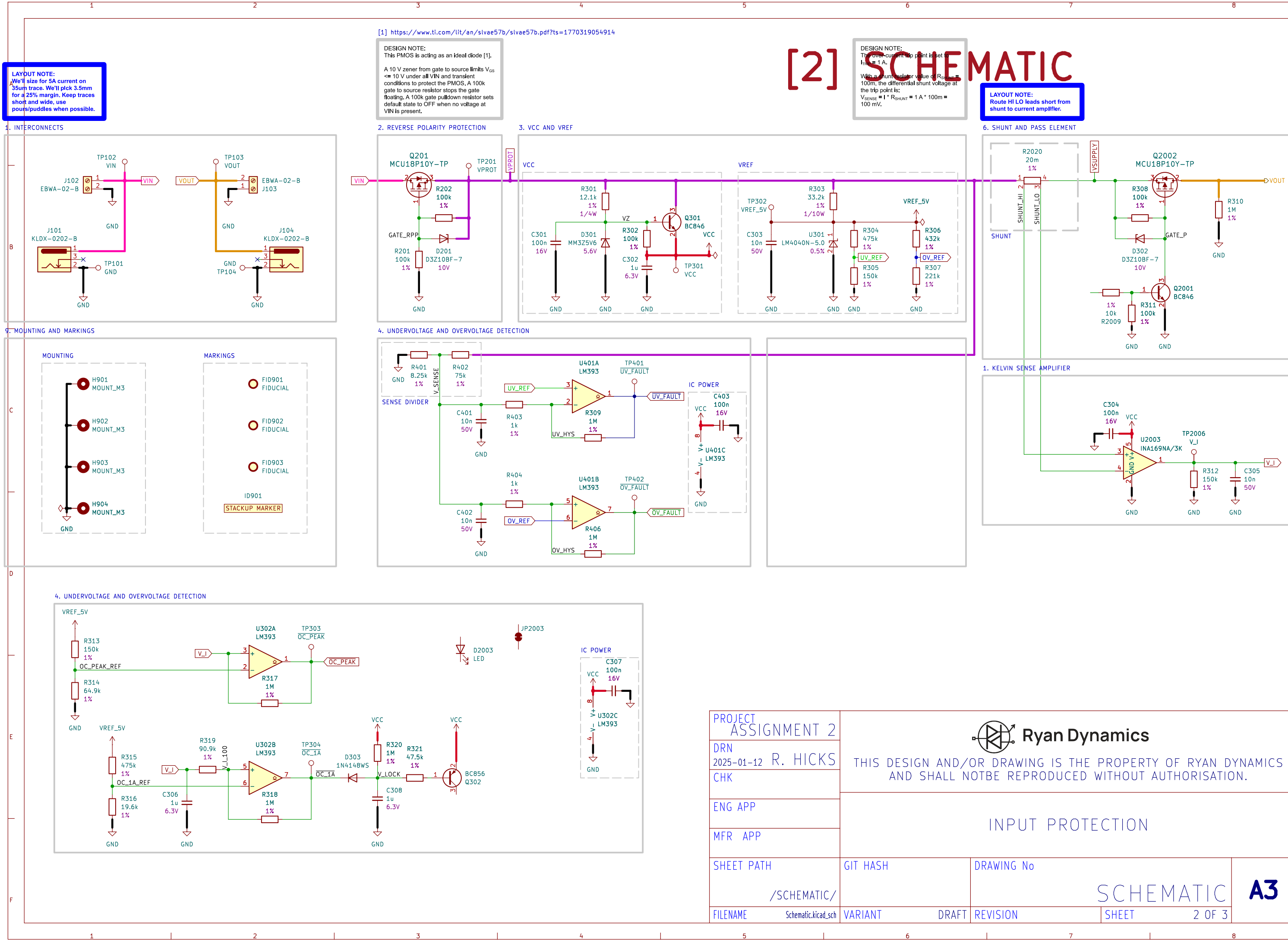
DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

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DRN 2025-01-12 R. HICKS						
CHK						
ENG APP		INPUT PROTECTION				
MFR APP						
SHEET PATH		GIT HASH		DRAWING No		A3
/						
FILENAME	Input-Protection.kicad_sch	VARIANT	DRAFT	REVISION	SHEET	1 OF 3



[1] <https://www.ti.com/lit/an/slvae57b/slvae57b.pdf?ts=1770319054914>

DESIGN NOTE:
This PMOS is acting as an ideal diode [1].

A 10 V zener from gate to source limits $V_{GS} \leq 10$ V under all VIN and transient conditions to protect the PMOS. A 100k gate to source resistor stops the gate floating. A 100k gate pulldown resistor sets default state to OFF when no voltage at VIN is present.

DESIGN NOTE:
The over-current trip point is set $I_{TRIP} = 1$ A.

With a shunt resistor value of $R_{SHUNT} = 100m$, the differential shunt voltage at the trip point is:
 $V_{SENSE} = I \cdot R_{SHUNT} = 1 A \cdot 100m = 100$ mV.

LAYOUT NOTE:
Route HI LO leads short from shunt to current amplifier.

2. REVERSE POLARITY PROTECTION

3. VCC AND VREF

6. SHUNT AND PASS ELEMENT

4. UNDERVOLTAGE AND OVERVOLTAGE DETECTION

1. KELVIN SENSE AMPLIFIER

4. UNDERVOLTAGE AND OVERVOLTAGE DETECTION

PROJECT ASSIGNMENT 2

DRN
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CHK

ENG APP

MFR APP

SHEET PATH

/SCHEMATIC/

FILENAME

Schematic.kicad_sch

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INPUT PROTECTION

GIT HASH	DRAWING No	SCHEMATIC A3
VARIANT	DRAFT	
REVISION	SHEET	2 OF 3

[3] REVISION HISTORY

PROJECT
ASSIGNMENT 2

DRN
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CHK

ENG APP

MFR APP

SHEET PATH

/REVISION HISTORY/

FILENAME Revision History.kicad_sch



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INPUT PROTECTION

GIT HASH

DRAWING No

REVISION HISTORY

A4

VARIANT

DRAFT

REVISION

SHEET

3 OF 3